

DRISHYAM

From Footage to Action

Testing Document

Unit, Integration, E2E & AI Evaluation

Document ID: DRISHYAM-TEST-001

Version: 1.0 | 13 September 2026

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1. Testing Strategy

Pyramid: many fast unit tests, solid integration tests for API + DB, fewer critical E2E flows, dedicated AI evaluation suite on labeled video.

2. Unit Tests

- Auth token generation/validation
- Permission checks for each role
- Alert state machine transitions
- Severity mapping
- Input validation helpers
- Temporal filter logic (AI side)

3. Integration / API Tests

- Login → access protected route
- Create site → add camera → list cameras
- Simulate internal event → alert created → appears in list
- Acknowledge flow updates status and audit log
- Cross-org access returns 403

4. E2E Critical Flows

- Full admin setup (org → site → camera → invite)
- Operator receives alert (mock event) → acknowledges
- Camera offline → status reflects
- Role restriction prevents unauthorized action

5. AI Evaluation

Maintain a labeled test set of Indian-context videos for weapon, violence, intrusion, loitering. Track precision, recall, F1, and latency. Gate model updates on regression thresholds. Include hard negatives (tools that look like weapons, sports, etc.).

6. Performance / Load

Load test API with concurrent alert creation and WebSocket connections. Measure edge pipeline FPS and latency under multiple streams.